III B.Tech – II Semester

Course Code	**ADVANCED STRUCTURAL ANALYSIS** **Professional Elective-III**	L	T	P	C
23A01605c		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the fundamental concepts of arches, including three-hinged and two-hinged arches, and analyze the effects of horizontal thrust, bending moment, normal thrust, and radial shear.
2. **Apply** the moment distribution method to analyze single-bay, single-story portal frames with and without side sway.
3. **Analyze** continuous beams and portal frames using Kani's Method, including cases with and without settlement of supports.
4. **Solve** structural problems using the flexibility method for continuous beams and single-bay portal frames, considering support settlements and side sway effects.
5. **Evaluate** the stiffness method for analyzing continuous beams and single-bay portal frames with and without side sway, ensuring structural stability and performance.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. **Explain** the behavior of three-hinged and two-hinged arches and**analyze** the effects of horizontal thrust, bending moment, normal thrust, and radial shear. (Bloom's Level: Understand - L2, Analyze - L4)
2. **Apply** the moment distribution method to**analyze** single-bay, single-story portal frames with and without side sway. (Bloom's Level: Apply - L3, Analyze - L4)
3. **Analyze** continuous beams and portal frames using Kani's Method, including cases with and without settlement of supports.
4. **Solve** structural problems using the flexibility method for continuous beams and single-bay portal frames, considering support settlements and side sway effects.
5. **Evaluate** the stiffness method for analyzing continuous beams and single-bay portal frames with and without side sway, ensuring structural stability and performance.

CO – PO Articulation Matrix

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	2	1	-	-	-	-	2	-	1	2	2
CO -2	3	3	2	2	2	-	-	-	-	2	-	2	2	2
CO -3	3	3	3	2	2	-	-	-	-	2	-	2	3	2
CO -4	3	2	3	3	3	-	-	-	-	2	1	2	3	3
CO -5	3	3	3	3	3	-	-	-	-	2	1	2	3	3

UNIT – I

ARCHES: Three Hinged and Two Hinged Arches, Elastic Theory of Arches– Eddy'S Theorem –Determination of Horizontal Thrust, Bending Moment, Normal Thrust and Radial Shear–Effect of Temperature-Determination of Horizontal Thrust Bending Moment, Normal Thrust and Radial Shear–Rib Shortening and Temperature Stresses.

UNIT – II

MOMENT DISTRIBUTION METHOD for FRAMES: -Analysis of Single Bay Single

Storey Portal Frame Including Sides Way–Substitute Frame Analysis By Two Cycle Method.		
UNIT – III		
KANI`S METHOD: - Analysis of Continuous Beams With and Without Settlement of Supports-Single Bay Single Storey Portal Frames With and Without Side Sway.		
UNIT – IV		
FLEXIBILITY METHODS:- Flexibility Methods- Introduction-Application to Continuous Beams Including Support Settlements—Analysis of Single Bay Single Storey Portal Frames Without and With Side Sway.		
UNIT – V		
Stiffness Methods: Stiffness Methods – Introduction – Application to Continuous Beams Including Support Settlements – Analysis of Single Bay Single Storey Portal Frames Without and With Side Sway.		
TEXT BOOKS:		
1. Analysis of structures by Vazrani&Ratwani– Khanna Publications. 2. Theory of structures by Ramamuratam, jain book depot , New Delhi.		
REFRENCE BOOKS:		
1. Structural analysis by R.S.Khurmi, S.Chand Publications, NewDelhi. 2. Basic Structural Analysis by K.U.Muthuetal .,I.K. International PublishingHousePvt.Ltd 3. Theoryof Structures by Gupta SP, GSPundit and R Gupta,Vol II, Tata McGraw Hill Publications Company td. 4. D. S. PrakashRao,“StructuralAnalysis:AUnifiedApproach”,UniversitiesPress		
Online Learning Resources:		
https://archive.nptel.ac.in/courses/105/106/105106050/		